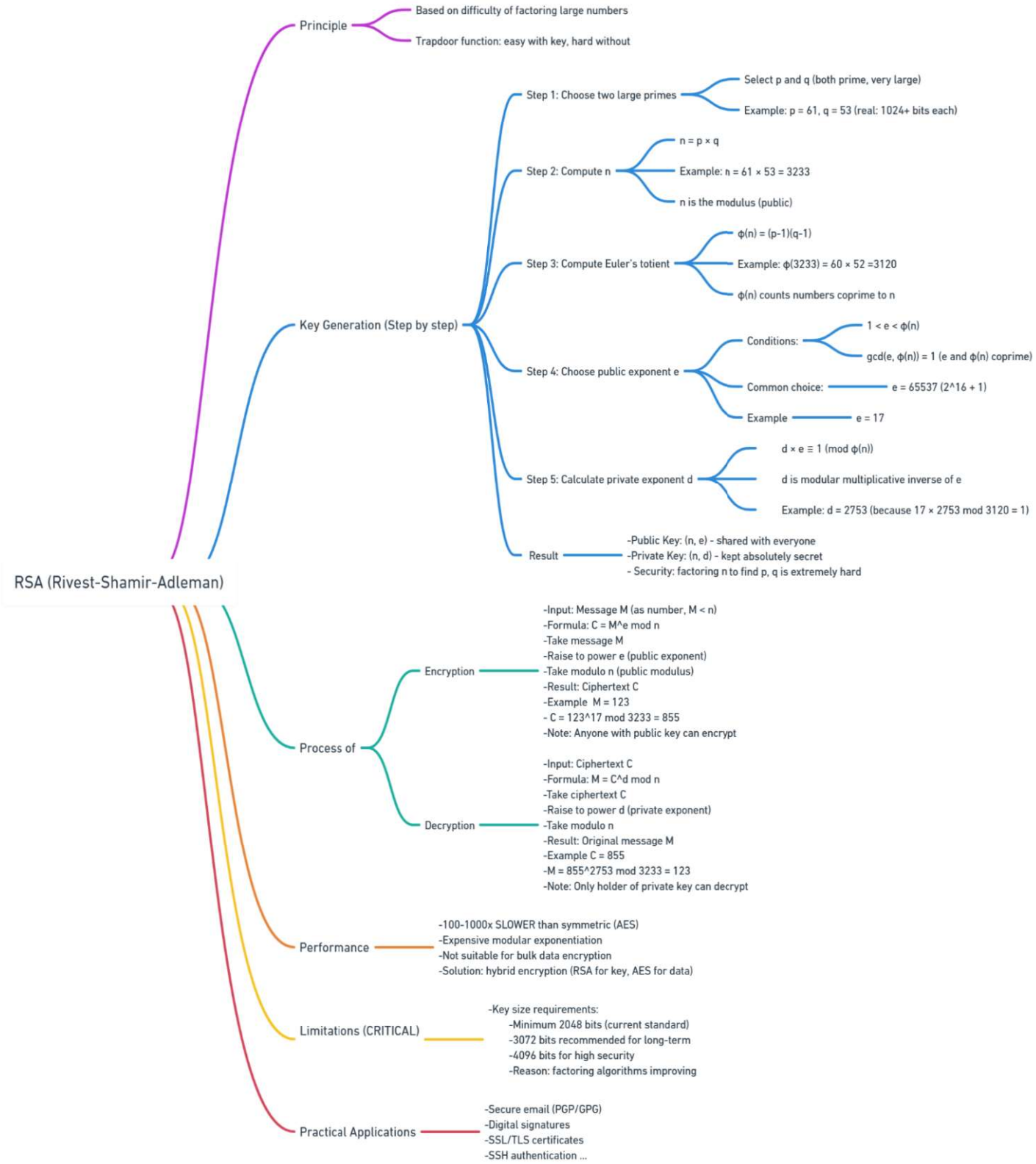
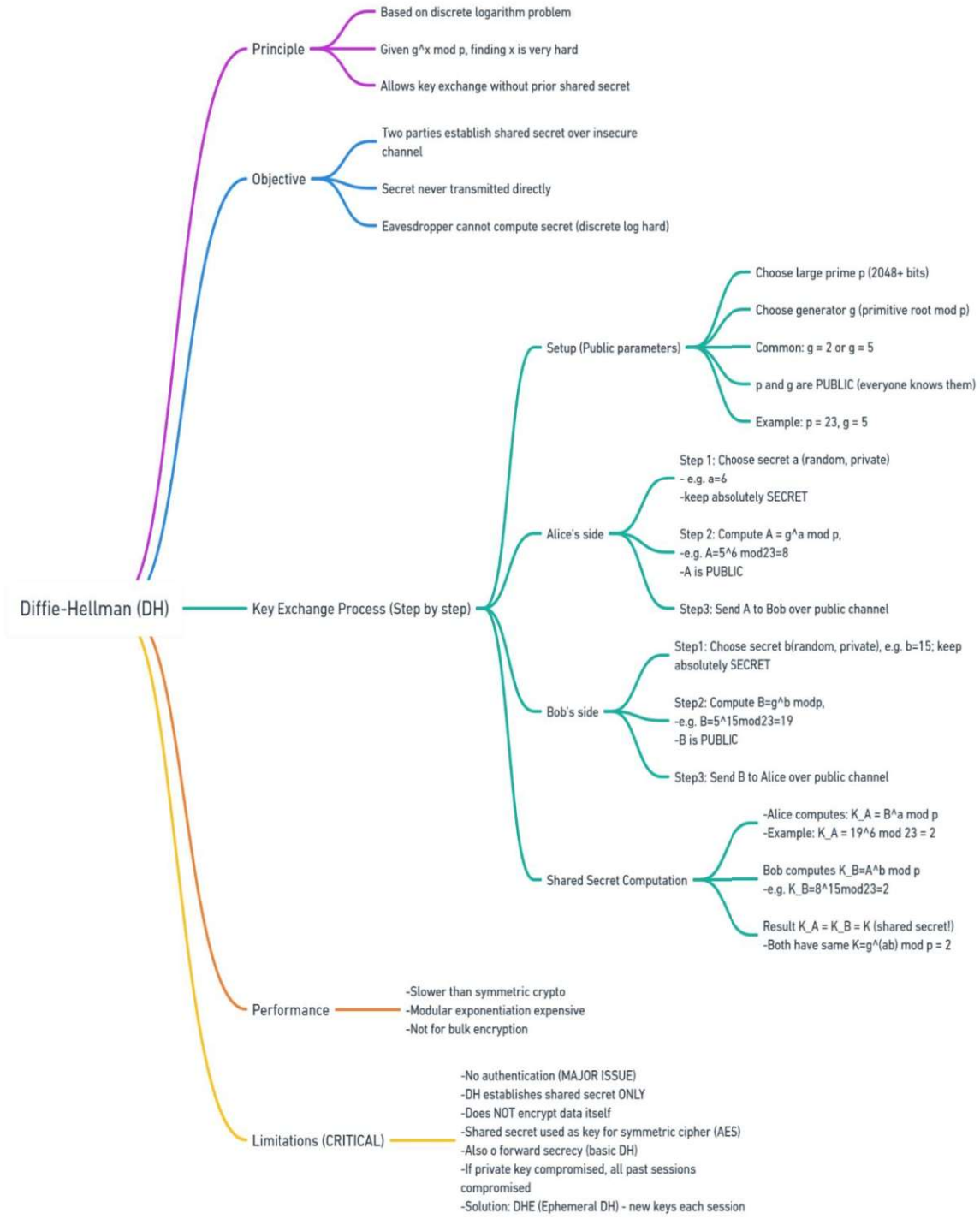
Major Asymmetric Systems

RSA :



DH :


Diffie-Hellman (DH)
Principle
Based on discrete logarithm problem
Given g^x mod p, finding x is very hard
Allows key exchange without prior shared secret
Objective
Two parties establish shared secret over insecure channel
Secret never transmitted directly
Eavesdropper cannot compute secret (discrete log hard)
Key Exchange Process (Step by step)
Setup (Public parameters)
Choose large prime p (2048+ bits)
Choose generator g (primitive root mod p)
Common: g = 2 or g = 5
p and g are PUBLIC (everyone knows them)
Example: p = 23, g = 5
Alice's side
Step 1: Choose secret a (random, private)
- e.g. a=6
-keep absolutely SECRET
Step 2: Compute A = g^a mod p,
-e.g. A=5^6 mod23=8
-A is PUBLIC
Step3: Send A to Bob over public channel
Bob's side
Step1: Choose secret b(random, private), e.g. b=15; keep absolutely SECRET
Step2: Compute B=g^b modp,
-e.g. B=5^15mod23=19
-B is PUBLIC
Step3: Send B to Alice over public channel
Shared Secret Computation
-Alice computes: K_A = B^a mod p
-Example: K_A = 19^6 mod 23 = 2
Bob computes K_B=A^b mod p
-e.g. K_B=8^15mod23=2
Result K_A = K_B = K (shared secret!)
-Both have same K=g^(ab) mod p = 2
Performance
-Slower than symmetric crypto
-Modular exponentiation expensive
-Not for bulk encryption
Limitations (CRITICAL)
-No authentication (MAJOR ISSUE)
-DH establishes shared secret ONLY
-Does NOT encrypt data itself
-Shared secret used as key for symmetric cipher (AES)
-Also o forward secrecy (basic DH)
-If private key compromised, all past sessions compromised
-Solution: DHE (Ephemeral DH) - new keys each session

ElGamal :



Comparison :

